# ABIDES Daily Investor Baselines and Parameter Sweep Results

Belief-Aware RL Project

April 27, 2026

## 1 Research Goal and Report Roadmap

The central question is whether an RL trader can make better decisions by estimating a latent belief over the market participants it is trading against. In a real market, the agent does not observe each participant's strategy directly. It only observes public market data such as prices, spreads, order book imbalance, volume, and recent returns. The goal is therefore not to identify every individual trader, but to infer enough about the hidden market composition to choose better actions.

Formally, we want to learn a belief-conditioned trading policy:

$$b_t = P(\text{market regime or participant mix} \mid \text{public history up to } t), \qquad a_t = \pi(o_t, b_t),$$

where $o_t$ is the usual market observation and $b_t$ is an online estimate of whether the current market is more noise-dominated, momentum-dominated, or mean-reversion-dominated. A standard RL agent only sees $o_t$. A belief-aware agent sees both $o_t$ and $b_t$.

### 1.1 What Are We Trying To Do?

We are trying to test whether public market data contains enough information to infer useful hidden structure about the agents generating the market. If the market is dominated by momentum traders, then following short-horizon pressure may be profitable. If the market is dominated by value or mean-reversion traders, then fading short-term moves may be better. A fixed policy averages across these regimes. A belief-aware policy should adapt as its estimate of the current regime changes.

### 1.2 What Experiments Are We Running?

This report studies two preliminary experiments that set up the belief-aware RL question.

First, we run a fixed baseline benchmark in the ABIDES daily-investor environment. Six hand-coded strategies trade in the same AAPL-calibrated market: `HOLD`, `BUY_HOLD`, `RANDOM`, `MOMENTUM`, `MR`, and `MR_INV`. This tells us which simple signals the default market rewards.

Second, we run one-at-a-time parameter sweeps over the background market and environment settings. These sweeps change the number and strength of value agents, momentum agents, noise agents, market-maker settings, order size, and timestep duration. The purpose is to check whether changing hidden market composition changes which strategy performs best.

### 1.3 How Do These Results Help?

The baseline and sweep results are not yet the final belief-aware RL experiment. They are the diagnostic layer that tells us whether belief estimation is worth doing. If every market composition

rewarded the same strategy, then estimating the hidden participant mix would add little value. But if different compositions favor different behavior, then a belief module has a real signal to exploit.

The current results support that motivation. The default AAPL calibration is strongly momentum-friendly: the `MOMENTUM` heuristic outperforms the mean-reversion heuristics by a wide margin. The value-agent sweep then shows the direction we care about: increasing fundamental anchoring makes mean-reversion more competitive and weakens momentum. This is exactly the kind of regime dependence a belief-aware policy should learn to detect from public history.

### 1.4 What Are The Next Steps?

The next step is to replace hand-coded strategy selection with a learned belief-conditioned policy. Concretely, we should:

1. Define a small set of latent market regimes, such as noise-dominated, momentum-dominated, value-agent-dominated, and mixed.
2. Generate ABIDES training data from each regime while exposing only public market observations to the learning agent.
3. Train a belief module that maps recent public history to a regime belief vector.
4. Train and compare two RL traders: a baseline policy $\pi(o_t)$, and a belief-aware policy $\pi(o_t, b_t)$.
5. Evaluate both trading performance and belief quality using P&L, Sharpe ratio, drawdown, win rate, and belief accuracy against the known simulated regime.

The rest of this document follows that flow. The setup sections explain the ABIDES market, the trading interface, and the heuristic policies. The results sections show which policies work under the default calibration and how those rankings shift under parameter sweeps. The conclusion translates those findings into the next belief-aware RL implementation plan.

## 2 Experimental Setup

The experiment evaluates the `markets-daily_investor-v0` ABIDES-Gym environment under an AAPL-calibrated `rmsc04` background market. The daily investor chooses among three actions at each step: market buy, hold, or market sell. Performance is measured by final mark-to-market P&L relative to the starting cash account.

The baseline benchmark uses 50 episodes per strategy with a 60 second timestep, starting cash of 1,000,000 cents, and fixed order size of 10 shares. All P&L numbers below are in cents.

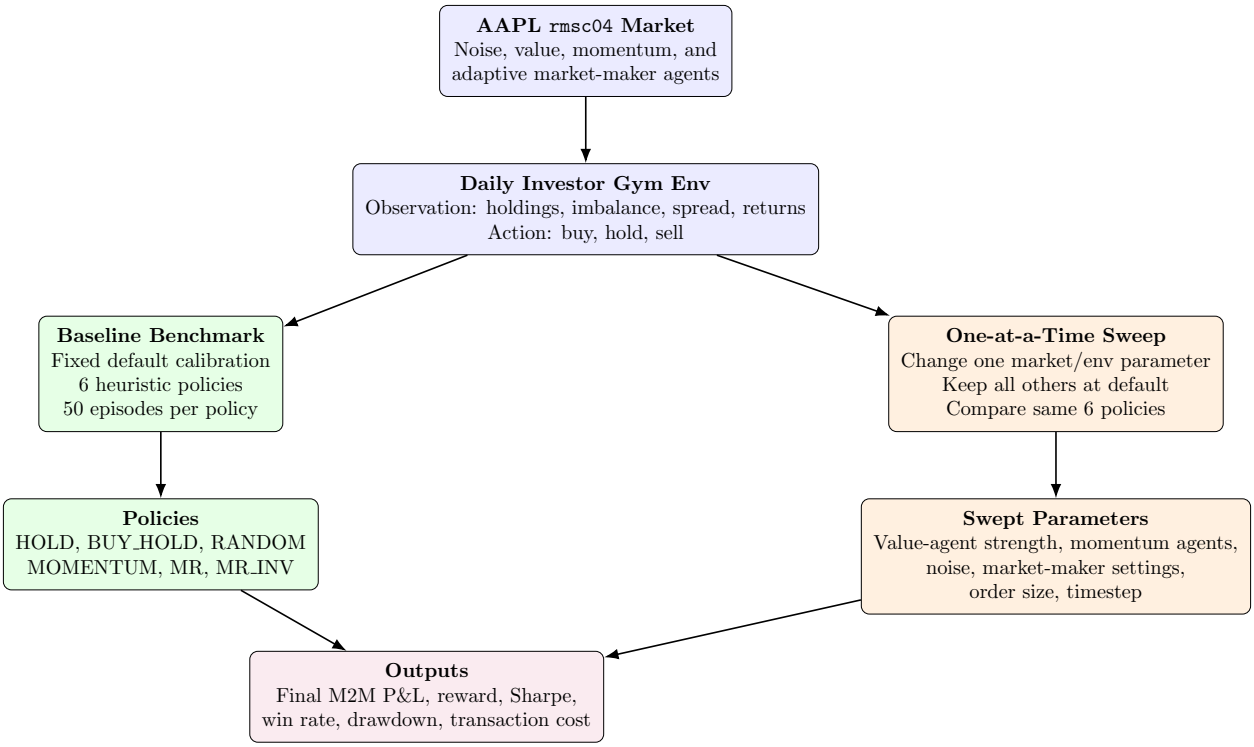



Figure 1: Experimental design. The baseline benchmark evaluates six fixed heuristics under the default AAPL calibration. The sweep keeps the same policy set but changes one environment or background-market parameter at a time to identify which market regimes favor momentum or mean reversion.

## 3 Setup Context

The daily investor is not trading against a simple price process. It trades inside an agent-based limit-order-book simulation. The background agents create order flow, liquidity, and price movement; the daily investor observes the resulting market state and submits one market order decision per environment step.

### 3.1 Background Market Agents

| Agent type | Role in the simulated market |
|---|---|
| Noise agents | Submit random buy or sell orders. They create uninformed order flow and volume, which affects book depth, spread costs, and short-horizon price noise. |
| Value agents | Trade when the market price deviates from their belief about fundamental value. More active value agents tend to anchor prices and make the market more mean-reverting. |
| Momentum agents | Trade with short-horizon trends, typically buying after upward movement and selling after downward movement. More momentum activity can make price moves persist instead of immediately reverting. |
| Adaptive market makers | Continuously provide bid and ask liquidity around the current market. Their wake-up frequency, spread rule, and sizing determine how costly it is for the daily investor to cross the spread. |

Table 1: Background agents in the `rmsc04` market configuration.

### 3.2 Daily Investor Interface

At each step, the daily investor receives a compact observation vector derived from the order book and its own portfolio, then chooses one of three actions.

| Quantity | Location | Meaning |
|---|---|---|
| Holdings | `obs[0]` | Current signed share inventory. Positive means long, negative means short. |
| Imbalance | `obs[1]` | Bid volume divided by total visible bid and ask volume. Values above 0.5 indicate stronger bid-side pressure. |
| Spread | `obs[2]` | Best ask minus best bid, in cents. This is the immediate cost paid by market orders. |
| Direction feature | `obs[3]` | Difference between mid price and last transaction price, a short-horizon microstructure signal. |
| Recent returns | `obs[4:7]` | Lagged mid-price returns used by the momentum and mean-reversion heuristics. |

Table 2: Observation features used by the daily investor environment.

The action space is deliberately simple: `0` submits a market buy, `1` holds, and `2` submits a market sell. Because these are market orders, active strategies pay spread and slippage whenever they trade. `HOLD` is therefore the zero-trading control: it should have exactly zero P&L in this setup.

### 3.3 Baseline Policies

The baseline policies are simple hand-coded trading rules. They are not trained RL agents; they are reference points that show which types of signals the market currently rewards.

| Policy | Trading rule and interpretation |
|---|---|
| `HOLD` | Never trades. This is the sanity-check baseline: final P&L should remain zero because cash and holdings never change. |
| `BUY_HOLD` | Buys once at the start and then holds. This measures whether the simulated day has a directional long bias. |
| `RANDOM` | Chooses buy, hold, or sell uniformly at random. This is a noisy lower bound for active trading because it pays transaction costs without using a signal. |
| `MOMENTUM` | Follows order-book pressure. If bid-side imbalance is high, it buys; if ask-side pressure is high, it sells. The hypothesis is that current order-book pressure predicts near-future price movement in the same direction. |
| `MR` | Mean reversion. It fades recent mid-price returns: after a price drop it buys, and after a price rise it sells. The hypothesis is that short-horizon moves overshoot and then revert. |
| `MR_INV` | Mean reversion with an inventory cap. It uses the same signal as `MR` but stops adding to positions once inventory gets too large. This tests whether mean-reversion losses are caused mainly by uncontrolled position buildup. |

Table 3: Hand-coded baseline policies evaluated in both the fixed benchmark and the parameter sweep.

### 3.4 How To Interpret the Sweep

The sweep changes one parameter at a time while keeping all other parameters at their default AAPL calibration. This isolates directional effects: for example, increasing `num_value_agents` tests whether stronger fundamental anchoring makes mean reversion more profitable, while increasing `num_momentum_agents` tests whether trend-following background flow makes momentum more profitable. Since the sweep plots show mean P&L with one standard deviation error bars, wide overlapping intervals should be interpreted as high uncertainty rather than a decisive ranking.

## 4 Baseline Strategy Results

| Strategy | Mean P&L | Std. P&L | Sharpe | Win Rate | Mean DD | Txn. Cost |
|---|---|---|---|---|---|---|
| HOLD | 0.00 | 0.00 | 0.000 | 100.0% | 0.000 | 0.00 |
| BUY_HOLD | -643.20 | 7336.97 | -0.088 | 46.0% | 0.009 | 1.25 |
| RANDOM | 6033.86 | 63751.45 | 0.095 | 58.0% | 0.072 | 351.80 |
| MOMENTUM | 66171.66 | 131290.71 | 0.504 | 82.0% | 0.122 | 371.30 |
| MR | -57942.00 | 120705.01 | -0.480 | 36.0% | 0.140 | 462.57 |
| MR_INV | -54810.98 | 26958.35 | -2.033 | 2.0% | 0.062 | 375.60 |

Table 4: Baseline performance over 50 episodes. Sharpe is computed as mean final P&L divided by standard deviation of final P&L across episodes.

The baseline result is clear: the default market regime is momentum-friendly and not mean-reversion-friendly. `MOMENTUM` has the highest mean P&L, the best Sharpe ratio, and an 82% win rate. The mean-reversion policies lose money on average. `MR_INV` reduces variance and drawdown relative to plain `MR`, but it is still consistently negative, with only a 2% win rate.

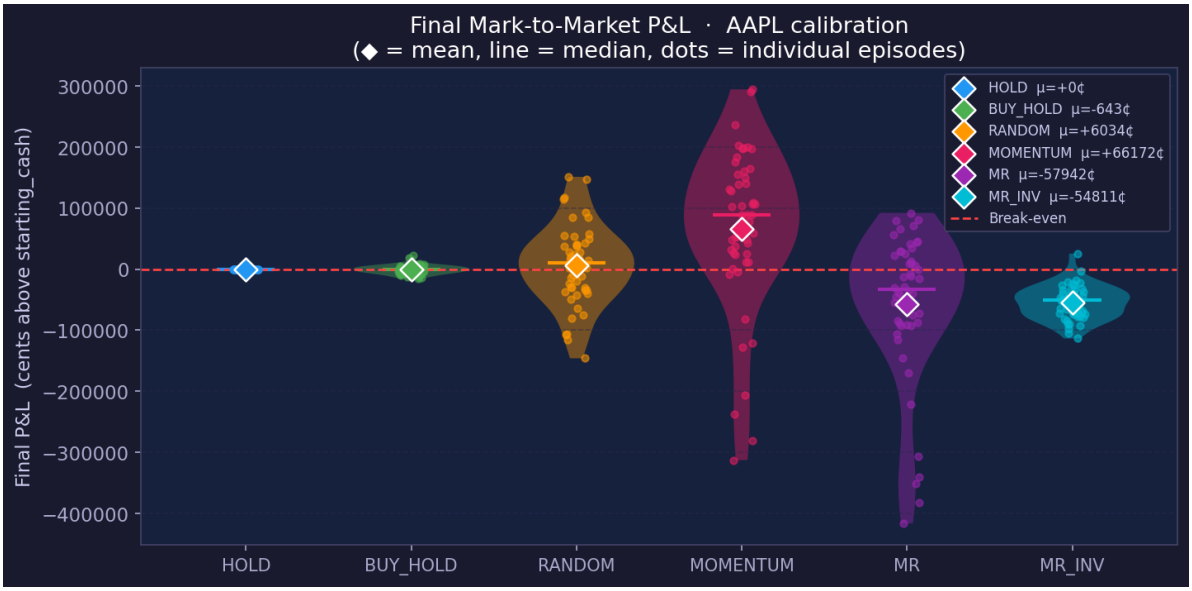
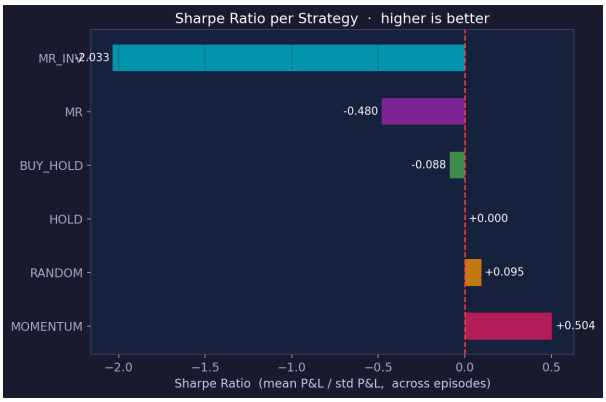
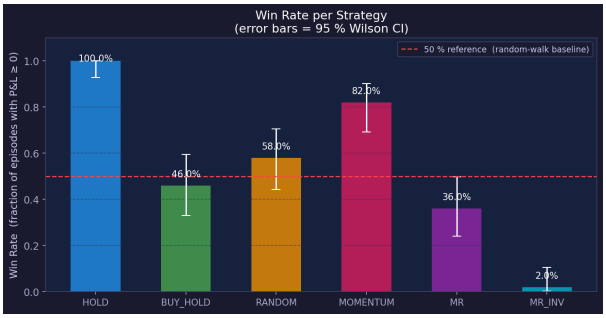



Figure 2: Final mark-to-market P&L distribution. Diamonds denote means, median lines show central tendency, and dots are individual episodes.

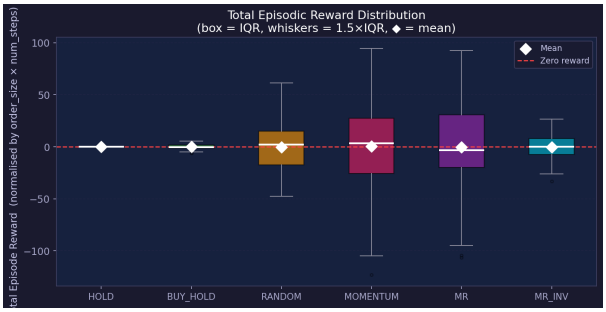



(a) Sharpe ratio

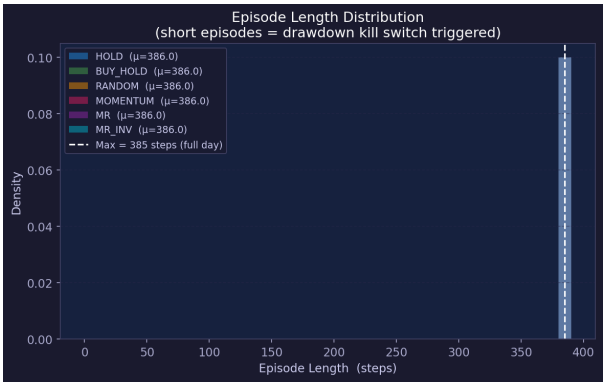



(b) Win rate

Figure 3: Risk-adjusted and directional success metrics.




(a) Episode reward




(b) Episode length

Figure 4: Reward and termination behavior. All strategies run for the full day, so the drawdown kill switch is not the main driver of the results.

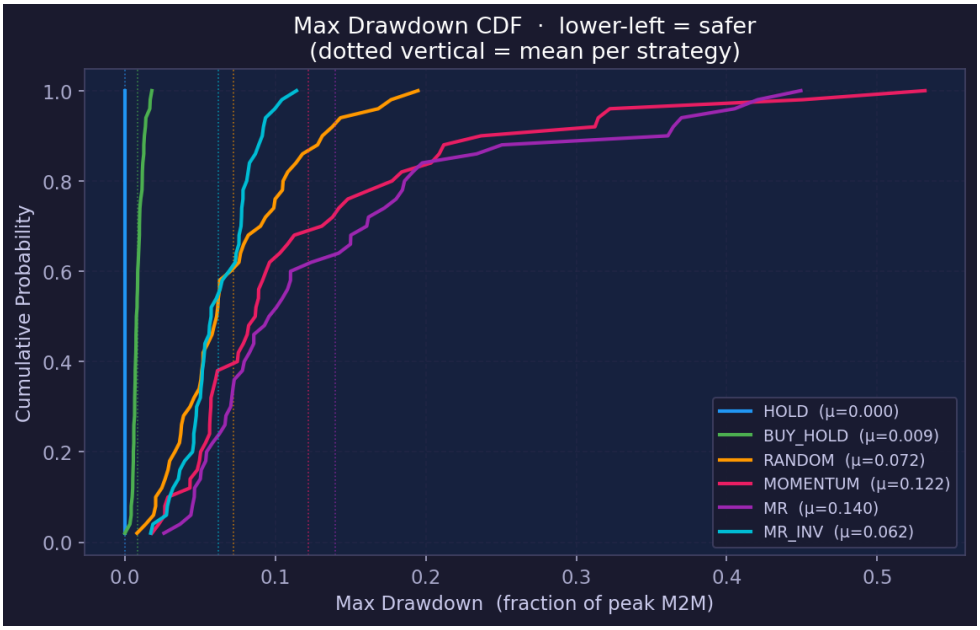



Figure 5: Maximum drawdown CDF. Curves farther left are safer. `MR_INV` has materially lower drawdown than `MR`, but its P&L remains negative.

## 5 Parameter Sweep Results

The sweep varies one parameter at a time and compares the same baseline strategies. The available plots show mean final P&L with one standard deviation error bars. The raw `sweep_results.json` file was not present alongside the plots, so these conclusions should be read as qualitative interpretations of the figures rather than exact statistical estimates.

| Sweep | Main interpretation |
|---|---|
| `num_value_agents` | The cleanest qualitative result. As value-agent count increases, `MR` improves and `MOMENTUM` weakens, consistent with stronger fundamental anchoring creating a more mean-reverting market. |
| `num_momentum_agents` | The expected monotonic benefit to `MOMENTUM` is not clean in the plot; variance is large and most bars overlap zero. |
| `lambda_a` | Higher value-agent arrival rate appears favorable to `MR`, but the standard deviations are large. |
| `kappa`, `kappa_oracle` | Mean-reversion strength changes strategy rankings, but the plots are too noisy for a strong one-parameter claim. |
| `mm_wake_up_freq` | The 60s setting looks most favorable to `MR`; 120s is worse for `MOMENTUM`. Error bars remain wide. |
| `order_fixed_size`, `timestep_duration` | No stable monotonic transaction-cost story is visible from the current sweep. |
| `fund_vol`, `num_noise_agents`, market-maker parameters | These change opportunity and noise, but current sweep variance is large enough that results should be treated as directional only. |

Table 5: Qualitative interpretation of the one-at-a-time parameter sweep plots.

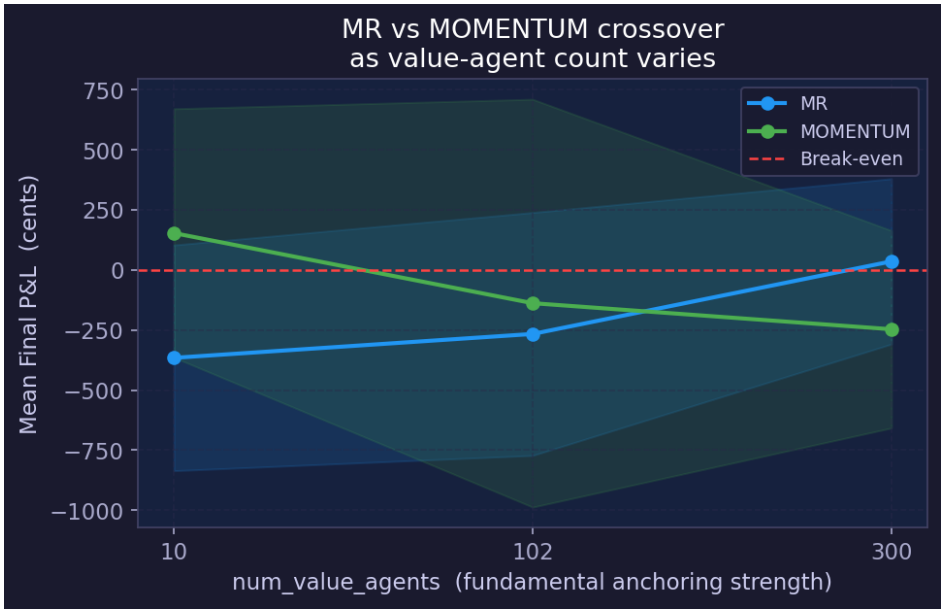
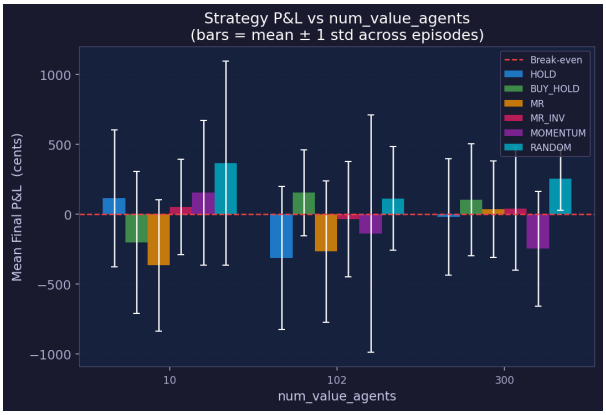
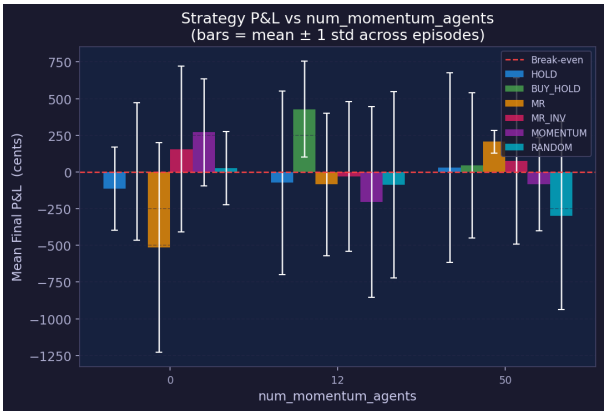



Figure 6: `MR` versus `MOMENTUM` as value-agent count varies. This is the most important sweep figure: more value agents move the market toward a mean-reverting regime.

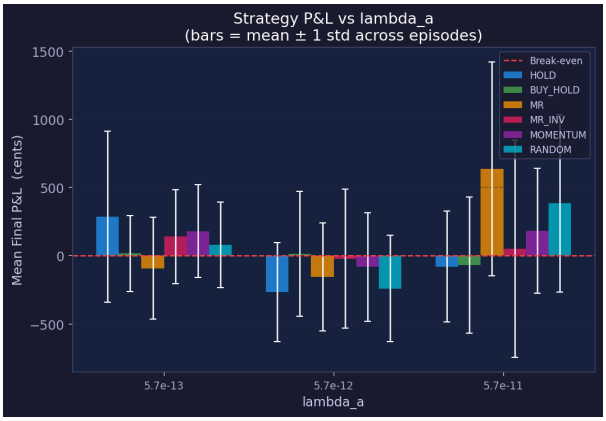



(a) Value agents

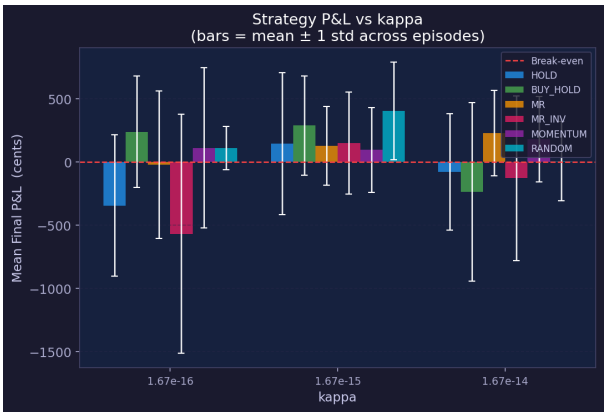



(b) Momentum agents

Figure 7: Agent-population sweeps.

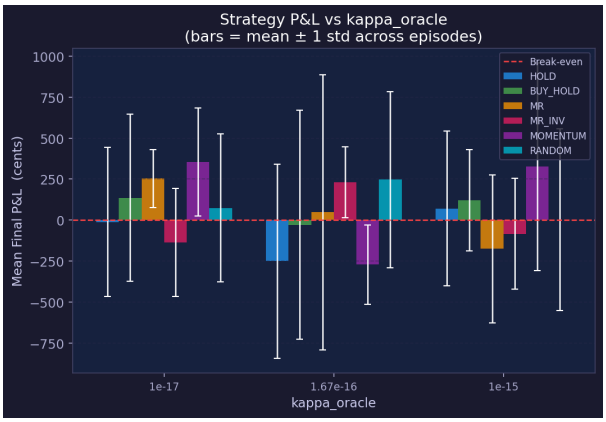



(a) `lambda_a`

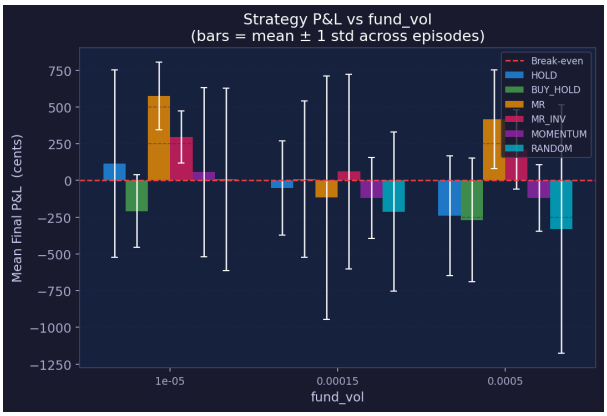



(b) `kappa`

Figure 8: Value-agent belief and arrival sweeps.

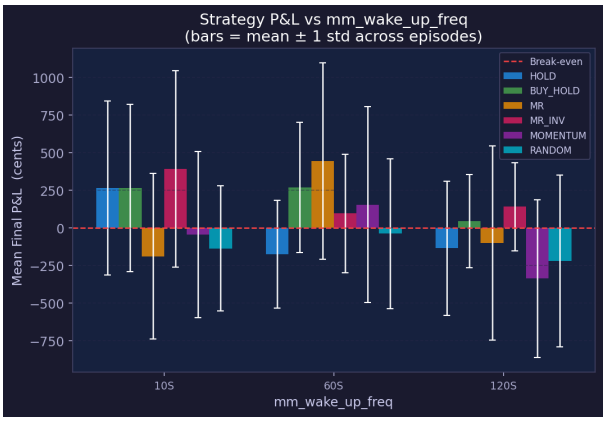



(a) `kappa_oracle`

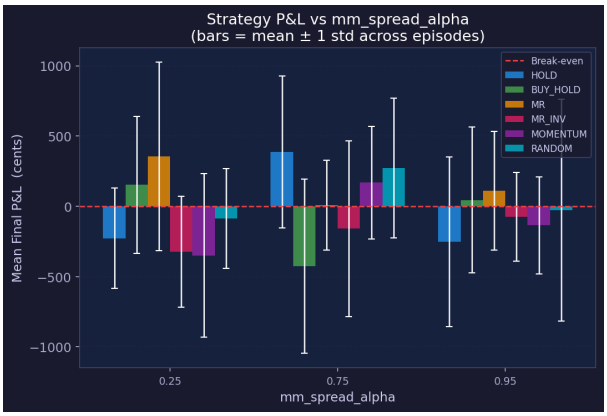



(b) `fund_vol`

Figure 9: Oracle fundamental-process sweeps.

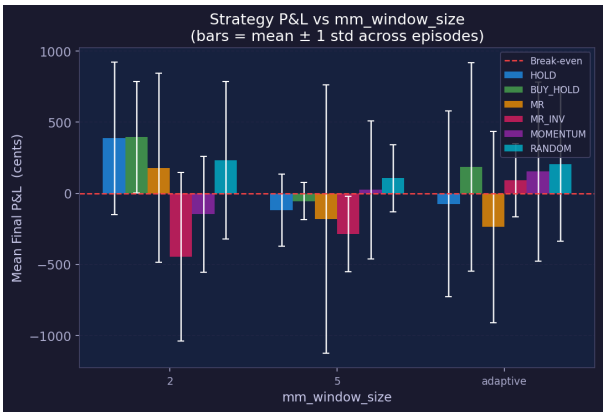



(a) Wake-up frequency

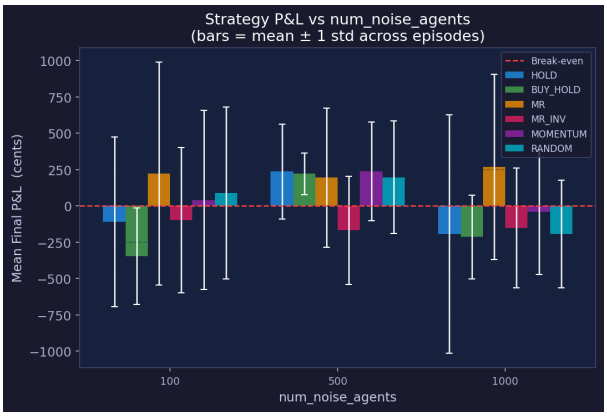



(b) Spread alpha

Figure 10: Market-maker timing and spread-adaptation sweeps.



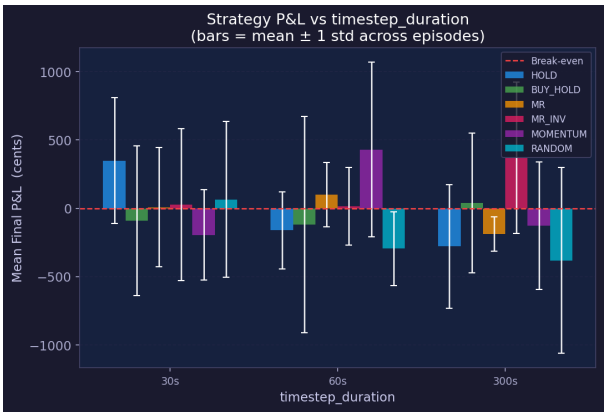

(a) Window size




(b) Noise agents

Figure 11: Liquidity and noise sweeps.

(a) Order size




(b) Timestep duration

Figure 12: Environment-level action-frequency and order-size sweeps.

## 6 Takeaways

The default AAPL-calibrated `rmsc04` regime rewards short-horizon momentum more than mean reversion. This matters for belief-aware RL training: if the goal is to test whether an agent can infer and exploit fundamental anchoring, the environment should likely be shifted toward stronger value-agent activity, for example by increasing `num_value_agents` or `lambda_a`. If the goal is to learn robust trading under the current default market, `MOMENTUM` is the benchmark to beat, not `MR`.

The sweep figures are useful for selecting candidate regimes, but most parameter effects need more episodes or the raw sweep table before they should be treated as statistically reliable.